

EE 595: Photonic Computing

Course Code: EE 595

Course Name: Photonic Computing

L-T-P-C: 2-0-2-3

Prerequisites: Either of EE 205, EP 302, EE 311, CS 201, CS 208, EE 551, or PH 502.

Intended for: UG, PG, PhD

Mutual Exclusion: None.

Approval: 65th BoA (approved with minor revisions)

Preamble

As conventional electronic scaling faces fundamental limits, photonics offers a viable pathway for accelerating compute and reducing data movement bottlenecks. Photonic computing is emerging as a promising paradigm due to its potential for high bandwidth, inherent parallelism, and improved energy efficiency compared to electronic systems, particularly for data-intensive applications. This course introduces photonic computing with a primary emphasis on silicon photonics and photonic integrated circuits (PICs) as scalable platforms for next-generation computation. Designed for a diverse audience, the course begins with foundational concepts in computing and photonics before developing device level understanding of integrated photonic components. Students learn how interference and propagation in waveguides enable linear algebra operations central to modern workloads such as artificial intelligence. The course focuses on interferometric meshes, wavelength-division multiplexing, and electro-photonic co-design, while briefly introducing alternative platforms for context. Through simulation-driven laboratories, students gain the ability to design and evaluate PIC-based photonic computing systems.

Course Modules

- **Unit 1: Motivation and Computational Limits:** Moore's law limitations, Von Neumann bottleneck, AI workloads and linear algebra dominance, Energy and data movement constraints, General purpose vs. specific purpose computing (2 hours)
- **Unit 2: Foundations of Photonics and Integrated Platforms:** Foundational optical principles (Maxwell's equations, Fresnel Laws), Guided-wave optics and confinement, Waveguide fundamentals, Silicon photonics platform overview, CMOS compatibility and fabrication basics. (4 hours)
- **Unit 3: PIC Components and Devices:** strip, rib, slot, Directional couplers and splitters, Mach-Zehnder interferometers (MZIs), Phase shifters (thermal, carrier-based), Modulators and photodetectors, Loss mechanisms and fabrication tolerances. (6 hours)
- **Unit 4: Photonic Computing Primitives:** Interference-based computation, Matrix-vector multiplication using MZI networks, Encoding: phase, amplitude, wavelength, Detection and readout, Noise, precision, and analog limits. (6 hours)
- **Unit 5: Integrated Photonic Architectures:** MZI mesh architectures (Reck, Clements), Programmability and calibration, WDM-based parallel computing, Time-multiplexed systems, Electronic-photonic co-design (DAC/ADC constraints), Benchmarking against conventional computing platforms. (6 hours)
- **Unit 6: Case Studies and Outlook:** Silicon photonic neural networks, Convolutional processing on-chip, Scaling challenges: thermal crosstalk, footprint, Packaging and fiber to-chip coupling, Brief survey of

alternative platforms (free space, diffractive, KANs, photonic reservoir computing, substrate computing, extreme machine learning), Application examples (RF, optical communications, data centers). (6 hours)

Laboratory/practical/tutorial Modules (10 hours)

- Waveguide mode simulation
- Directional coupler and splitter design
- MZI transfer function and tuning
- 2x2 and 4x4 interferometric mesh simulation
- Matrix multiplication using MZI mesh
- WDM-based parallel computation
- Noise and phase error analysis
- Mini Project: PIC-based system design

Text Books

1. Chrostowski, Lukas, and Michael Hochberg. *Silicon photonics design: from devices to systems*. Cambridge University Press, 2015.
2. Saleh, Bahaa EA, and Malvin Carl Teich. *Fundamentals of photonics, 2 volume set*. John Wiley & sons, 2019.

References

1. Y. Shen et al., Deep learning with coherent nanophotonic circuits, *Nature Photonics* (2017)
2. J. Feldmann et al., Parallel convolutional processing using integrated photonics, *Nature* (2021)
3. D. A. B. Miller, Attojoule optoelectronics, *JLT* (2017)
4. W. R. Clements et al., Optimal design of interferometers, *Optica* (2016)
5. B. J. Shastri et al., Photonics for AI, *Nature Photonics* (2021)
6. M. Atwany et al., A review of emerging trends in photonic deep learning accelerators, *Frontiers in Physics* (2024).
7. N. A. Mitchell et al., Photonic multiply-accumulate operations for neural networks, *IEEE Journal of Selected Topics in Quantum Electronics* (2019).